

Credit Card Default Prediction

Ranking risk, explaining patterns and choosing an action

NTU SC6122 Emerging Topics in FinTech — Group 5

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1. Problem, objectives and data

Credit default prediction can support the prioritization of account reviews. A missed default and an unnecessary review impose different costs, so the model with the best classification accuracy need not support the best decision. We ask: **how well can models identify default risk, and how should ranking ability, interpretability and decision cost be balanced?** We compare logistic regression (LR), a decision tree (DT), random forest (RF) and XGBoost on the same historical records. Objectives are to evaluate ranking using average precision (AP) and ROC-AUC, explain predictive associations, and examine threshold policies under explicit hypothetical error costs.

We use UCI's Default of Credit Card Clients dataset [1]: 30,000 Taiwan clients, 23 predictors and a binary default-payment outcome (1 = default). Predictors describe credit limit, age, sex, education, marital status, repayment status, bill amounts and previous payments. Payment history covers April–September 2005; monetary variables are in NT dollars. There are 6,636 positive outcomes (22.12%) and 23,364 negatives. Thus an always-negative classifier can look accurate while detecting no defaults. This is an educational historical-sample study, not evidence of readiness for current credit decisions.

Cleaning and identity. There are no missing entries. Undocumented EDUCATION values 0/5/6 are mapped to 4 (345 rows); MARRIAGE 0 is mapped to 3 (54 rows). Other values, including negative repayment codes and large or negative amounts, are retained. No rows, outliers or duplicate records are removed; 35 identical feature–label repetitions remain. Equal records alone do not establish repeated customers, but retaining them can create dependence across partitions. The final audit finds 15 test rows whose feature vectors also appear in development data. This limitation is retained rather than changing the established split after evaluation.

Traceability. The official UCI CSV was downloaded again during integration, and the above transformation reproduces the committed clean file exactly in values, types, row order and column order. The original download bytes were not saved; the new raw file and SHA256 audit are provided. The repository's zero-based `row_id` equals original UCI ID minus one. Neither identifier nor the label enters any model.

Partition	Clients	Use in the experiment
Fitting	18,000	Five-fold CV for model selection; selected model fitted on all fitting rows
Validation	6,000	Selection of decision thresholds
Test	6,000	Evaluation of frozen models and decisions

Fixed stratified partitions, seed 42. Each inner CV fold uses 14,400 training and 3,600 holdout rows.

2. Experimental design and model selection

The original 80/20 split yields 24,000 development and 6,000 test rows. A further stratified 75/25 development split produces 18,000 fit and 6,000 validation rows. Positive counts are 3982, 1,327 and 1,327 respectively. All four reproducible experiments have identical row-level development and five-fold CV memberships. IDs are unique and partitions disjoint. Final models fit only the 18,000 rows; validation rows are not incorporated by a later refit.

Each CV training fold fits its own preprocessing. SEX, EDUCATION and MARRIAGE are one-hot encoded with unknown categories ignored. LR additionally learns numeric StandardScaler statistics within the fold. Tree models retain numeric scales. No missing-value imputation or sampling is used. Selected class weights are part of the training configuration, not a change to test prevalence. Pipelines prevent categories or scaling moments from being learned on validation/test data.

Hyperparameters. The common selection objective is mean five-fold AP. Equal scores prefer the lowest candidate index. DT uses exhaustive grid search; RF/XGBoost evaluate a specified baseline plus 23 seeded random candidates. The harmonized LR run evaluates 12 exhaustive settings. These are comparable splits and objectives, not identical computational budgets.

Model	Configs	CV fits	Selected (summary)	CV AP
LR*	12	60	C=0.1; unweighted	0.5113
DT	21	105	Unlimited depth; leaf=100	0.5307
RF	24	120	500 trees; depth=8; leaf=2; class 1:3	0.5647
XGB	24	120	150 trees; depth=4; rate=.03; weight=3	0.5620

Table 1. Declared searches and selected configurations. The complete search spaces, per-fold scores, package versions and model hashes are in each `protocol_frozen.json` and `cv_results.csv`.

LR uses L2 logistic regression (`lbfgs`, maximum 3,000 iterations); its grid crosses C values .001, .01, .1, 1, 10 and 100 with no class weighting or balanced weighting. DT searches maximum depth 2, 3, 4, 5, 6, 8 or unlimited, and minimum leaf size 1, 20 or 100. RF varies tree count, depth, split/leaf size, feature subsampling and class weighting. XGBoost varies tree count, learning rate, depth, child weight, row/column sampling, L1/L2 penalties and positive-class weighting [3]. Baseline tree/ensemble configurations are retained alongside selected configurations.

Decision selection. We use $\hat{y} = 1[s \geq t]$ throughout, where s is the positive-class score. The fixed reference is $t = 0.5$. For LR, RF and XGBoost, all distinct validation score cutoffs are considered for $C_r = FP + r FN$, with $r \in \{1, 3, 5, 10\}$. Equal costs prefer the largest threshold (fewer alerts); tied scores move together. DT has only its historical fixed 0.5 policy. Each cost threshold is chosen on validation, then applied unchanged to test predictions. No test threshold sweep is used to select a policy.

LR contribution and comparison scope. Lei Peng contributed data inspection, preprocessing and the initial LR baseline; the team’s earlier LR result tables are retained in the repository. For the common comparison below, LR* denotes a separately logged integration run using the shared fitting rows, CV folds and metric implementation. Its 12-setting search was specified before that run, and its model and thresholds were frozen before test evaluation. This was a *post-hoc harmonization after earlier test results were available*, not a fresh independent holdout. The reported LR* values refer to that run, not a claim to reproduce the teammate’s original execution. All models retain the limitation of team-wide historical test exposure.

3. Ranking and classification findings

Model	AP	ROC-AUC	P	R	F1	Acc
LR* B	0.4970	0.7111	0.6975	0.2555	0.3740	0.8108
LR* T	0.4967	0.7107	0.6985	0.2532	0.3717	0.8107
DT B	0.2843	0.6076	0.3783	0.4039	0.3907	0.7213
DT T	0.5221	0.7538	0.6583	0.3572	0.4631	0.8168
RF B	0.5312	0.7534	0.6441	0.3806	0.4784	0.8165
RF T	0.5508	0.7752	0.5463	0.5200	0.5328	0.7983
XGB B	0.5544	0.7773	0.6565	0.3557	0.4614	0.8163
XGB T	0.5516	0.7762	0.5134	0.5486	0.5304	0.7852

Table 2. Historical test, $n = 6,000$, default positive. All rows use threshold 0.5. $P/R/F1/Acc$ are proportions. $B = \text{baseline}$; $T = \text{CV-selected}$; $LR^* = \text{harmonized integration run (Section 2)}$. Full precision and all confusion counts are in `model_comparison.csv`.

Model	B: TN	FP	FN	TP	T: TN	FP	FN	TP
LR*	4526	147	988	339	4528	145	991	336
DT	3792	881	791	536	4427	246	853	474
RF	4394	279	822	505	4100	573	637	690
XGB	4426	247	855	472	3983	690	599	728

Table 3. Test confusion counts at 0.5. $B = \text{baseline}$; $T = \text{CV-selected}$. Counts are recomputed from row-aligned scores and labels.

AP is computed by `average_precision_score`, a recall-increment weighted sum of precision, not trapezoidal PR-AUC [2]. ROC-AUC summarizes positive–negative score ordering. Both use continuous scores and are independent of the chosen operating threshold; precision, recall, F1 and accuracy describe a particular decision rule.

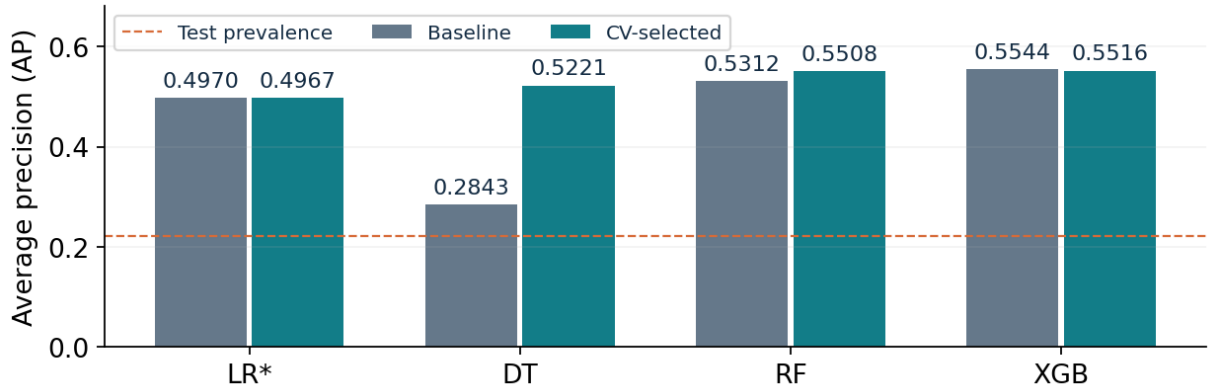


Figure 1. Test AP for declared baselines and CV-selected models. The dashed reference is positive prevalence (0.2212). Selection uses fit-fold CV, never this chart.

The unrestricted DT strongly overfits: mean fit-fold AP is about 1.0000 while CV AP is 0.2970. A minimum leaf size of 100 raises test AP from 0.2843 to 0.5221. RF selection raises AP from 0.5312 to 0.5508. XGBoost changes from 0.5544 to 0.5516: the selected configuration does not improve test ranking. LR* changes from 0.4970 to 0.4967, also without a ranking gain. Among the displayed point estimates the XGBoost baseline has the highest AP, but this retrospective observation does not authorize a new model-selection round. Small differences between RF and XGBoost do not establish superiority.

4. Thresholds, costs and review capacity

We treat one false-positive review as one hypothetical cost unit, and a missed default as r units. This is a sensitivity analysis, not a measured monetary loss: loan exposure, loss given default, treatment effects and review capacity are unavailable. Positive class weights can change score calibration, so a probability-derived cutoff such as $1/(1+r)$ is not assumed valid. Policies are selected empirically on validation data.

Model	Policy	Threshold	Recall %	Alert %	FP	FN	Cost
LR*	0.5	0.50000	25.32	8.02	145	991	5100
LR*	Val $r=5$	0.23432	59.38	32.75	1177	539	3872
RF	0.5	0.50000	52.00	21.05	573	637	3758
RF	Val $r=5$	0.32288	77.09	48.45	1884	304	3404
XGB	0.5	0.50000	54.86	23.63	690	599	3685
XGB	Val $r=5$	0.31502	82.52	55.72	2248	232	3408

Table 4. Same-model test comparison at 0.5 versus its validation-selected $r = 5$ policy. Alert rate is the share flagged for review; costs are total hypothetical units across 6,000 rows. DT has no historical validation-selected cost policy.

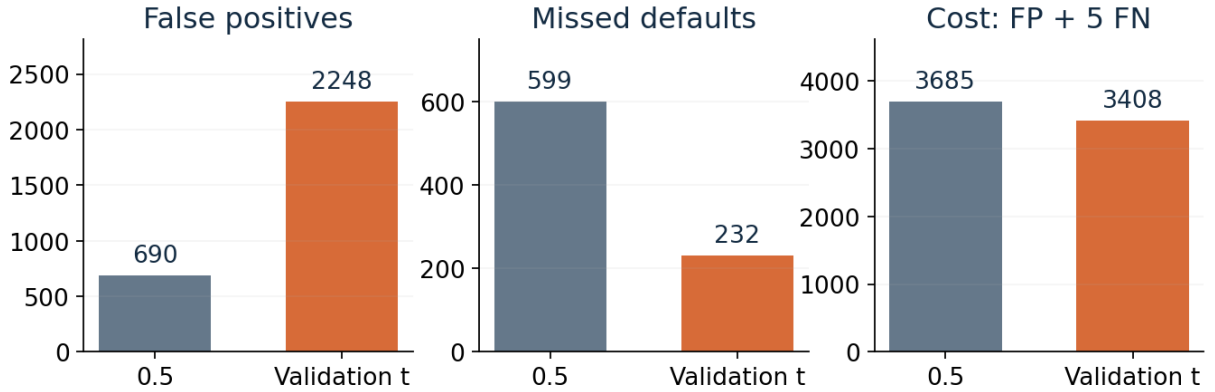


Figure 2. Moving the same tuned XGBoost from 0.5 to its validation cutoff changes FP/FN and workload. Ranking scores remain unchanged.

For XGBoost, $t = 0.31502$ changes cost from 3,685 to 3,408, a reduction of 7.52%. Recall rises from 54.86% to 82.52%, while false positives rise from 690 to 2,248 and 55.72% of clients are flagged. Precision falls from 51.34% to 32.76%. The benefit is fewer missed defaults under the assumed cost ratio; it is not a gain in AP or ROC-AUC. If that review load is infeasible, the policy would need a separately validated capacity constraint, not a threshold changed after viewing test errors.

RF at its validation $r = 5$ threshold costs 3,404 and flags 48.45%, versus XGBoost’s 3,408 and 55.72%. The four-unit difference is too small to support a strong cross-model cost claim. RF detects fewer defaults (77.09% recall) with fewer false positives. An always-negative policy costs 6,635 and an always-positive policy 4,673 at $r = 5$; both are useful context, not deployment recommendations.

Uncertainty. Historical 1,000-replicate stratified paired bootstraps condition on the frozen models, thresholds and fixed class counts. XGBoost’s tuned-minus-baseline AP difference is -0.0028 (95% interval $[-0.0106, 0.0054]$). The selected-minus-0.5 cost difference is -46.17 units per 1,000 (95% interval $[-74.34, -18.00]$). These intervals describe resampling of fixed predictions; they exclude retraining, hyperparameter/threshold selection, prevalence shifts and new-population uncertainty. They cannot be read as end-to-end deployment guarantees.

5. Interpretation and audit evidence

Complementary explanations. LR coefficients summarize regularized associations on standardized numeric and encoded categorical scales. Correlated bill/payment variables and full one-hot coding limit isolated coefficient interpretations. DT offers explicit paths: its root splits on `PAY_0 ≤ 1.5`; within fit rows, default proportions are 16.51% on the left and 70.13% on the right. These are node-level sample summaries, not causal effects or calibrated individual probabilities. The selected tree has 134 leaves and depth 18: no explicit depth cap does not mean no regularization, since every leaf requires at least 100 observations.

RF averages many trees, making full decision paths less concise. Its saved validation permutation importance identifies recent repayment status, `PAY_0`, as most influential: shuffling it reduces AP by 0.1971 on average, versus 0.0196 for `PAY_2`. Correlated variables can substitute for one another, and shuffling can create unrealistic combinations; importance describes predictive reliance, not causality. XGBoost instead builds regularized trees sequentially; its threshold trade-offs are evaluated in Section 4.

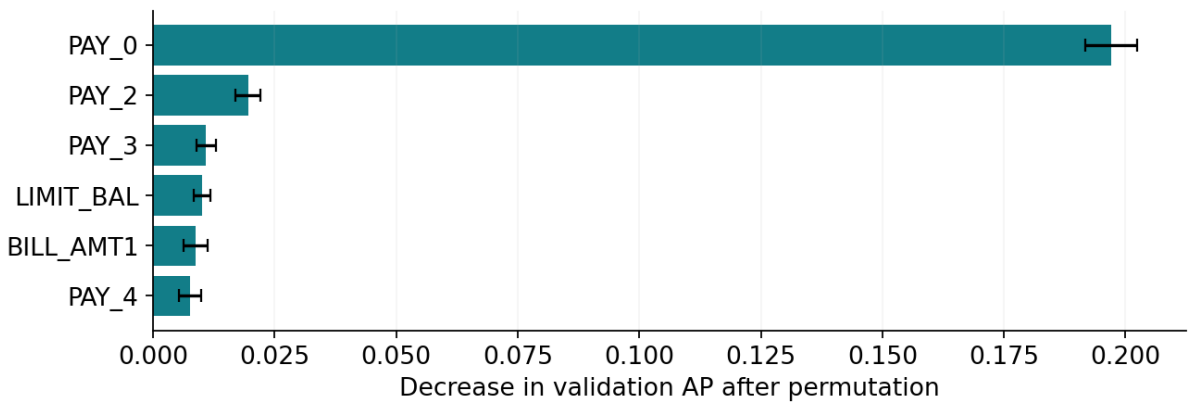


Figure 3. Tuned random-forest validation permutation importance, top six features. Error bars are the SD across five permutations, not confidence intervals or causal estimates. Horizontal units: decrease in validation AP.

Illustrative RF errors. At 0.5, the selected forest has 637 false negatives and 573 false positives. In the existing saved error export, row 982 defaults despite score 0.1012 and `PAY_0=-2`; row 28747 does not default despite score 0.9415 and `PAY_0=3`. These most-confident errors illustrate that repayment history is not deterministic; they are not representative average clients and are not used to retune the model. Both threshold-specific case exports satisfy their own classification rules. Extreme cases can remain errors under both policies even when overall error counts change.

Reconciliation. All metrics in the final comparison are recalculated after one-to-one `row_id` alignment with actual labels. CSV scores and thresholds use round-trip float parsing. Threshold-equality cases are explicitly tested with $\geq$; no rounding is performed before decisions. Source-train/test hash differences in historical protocols are explained by LF/CRLF serialization, while parsed content, types and order agree. Development and CV membership files are identical across the reproducible models.

Saved models reproduce stored decisions. XGBoost probability replay on this platform differs by at most 5.96×10^{-8} (float32-scale drift) while frozen policy decisions agree; original probabilities remain the reporting source. Model hashes and run metadata are checked, with newline-aware evidence for textual artifacts [4]. Audit checks establish reproducible evidence consistent with the code and artifacts; they cannot prove everything a teammate did outside the repository.

6. Conclusions, limitations and contributions

The historical sample contains useful predictive structure. A regularized tree improves on its unrestricted baseline and provides inspectable rules. RF and XGBoost obtain similar selected-model AP, while LR supplies an interpretable linear reference under the common comparison protocol.

Ranking quality and action policy answer different questions. XGBoost tuning did not yield a clear AP gain, but its validation-selected cutoff reduced hypothetical cost by accepting more false positives. Operational selection needs credible error costs and review capacity; test outcomes cannot justify another optimization round.

Limitations include one historical geography/time period; a random rather than temporal or external holdout; retained duplicate feature vectors; reuse of a previously examined test sample; unequal search budgets; unvalidated calibration; and no assessment of group-specific errors or fairness. Demographic fields warrant further governance evaluation. Cost ratios do not capture exposure, realized losses, intervention benefits or review limits. The conditional bootstrap omits the uncertainty from building and selecting models. We make no causal, regulatory-compliance or production-readiness claim.

A defensible next study would freeze a new protocol on fresh temporal/external data, assess calibration and subgroup errors, collect realistic loss and intervention-cost estimates, and evaluate policies constrained by review capacity. These are future steps, not experiments silently performed on the already evaluated test set.

Member contribution statement

Member / student ID	Responsibility and presentation	Share
Lei Peng G2509090C	Part 1: Data checks and preprocessing; logistic baseline; data characteristics and baseline performance; slides 1–3	25%
Zhang Hanyu G2509091L	Part 2: Decision-tree tuning; learned rules and overfitting control; slides 4–6	25%
Zhou Xinzhe G2509033F	Part 3: Random-forest tuning; feature importance and illustrative misclassification cases; slides 7–9	25%
Xu Yiqun G2509092H	Part 4: XGBoost tuning; threshold trade-offs between missed defaults and false alarms; slides 10–12	25%

Each member has an equal 25% contribution share. Shared responsibilities cover the data split, preprocessing and metrics, integration of the four sections, result checking and presentation preparation.

AI disclosure

Generative AI was used for language polishing and grammar guidance, and to assist with code development, experiment verification, analysis, and preparation of the report and presentation. The group is responsible for the submitted work, its interpretation and the appropriate acknowledgement of sources.

References

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- [2] scikit-learn developers. *average_precision_score* documentation. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html. Accessed 7 September 2026; experiments use scikit-learn 1.8.0.
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- [4] Group 5. *SC6122 Credit Default repository*, member experiment protocols and integration evidence, 2026. [Repository and audit artifacts](#).